Time Series HW4

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1 Task 1

1.1 Introduction

Financial markets may contain information about future price developments. We check whether the growth rate of the Dow Jones Industrial Average (DJIA) helps explain monthly US inflation. To check this we use ARDL and ARMAX.

1.2 Data Description

The data are obtained from the file `DJA.xls`, containing monthly observations of the Consumer Price Index (CPI) and the Dow Jones Industrial Average (DJIA). Both series are already expressed in natural logarithms. The sample spans from February 1900 to March 2019, yielding 1430 observations after taking first differences.

Let cpi_t denote the log CPI and $djia_t$ the log DJIA. Inflation and stock market growth are defined as:

$$\pi_t = cpi_t - cpi_{t-1}, \quad g_t = djia_t - djia_{t-1}.$$

1.3 Stationarity Analysis

Before modelling, the stationarity properties of the series are examined using the Augmented Dickey-Fuller (ADF) test and the KPSS test. The results are summarised in Table 1.

The ADF test fails to reject the unit root hypothesis for the log levels, while the KPSS test rejects stationarity at the 1% level. In contrast, both tests confirm that the first differences π_t and g_t are stationary (ADF p-values < 0.01, KPSS p-values > 0.05). Hence, all subsequent modelling is performed on these stationary transformations.

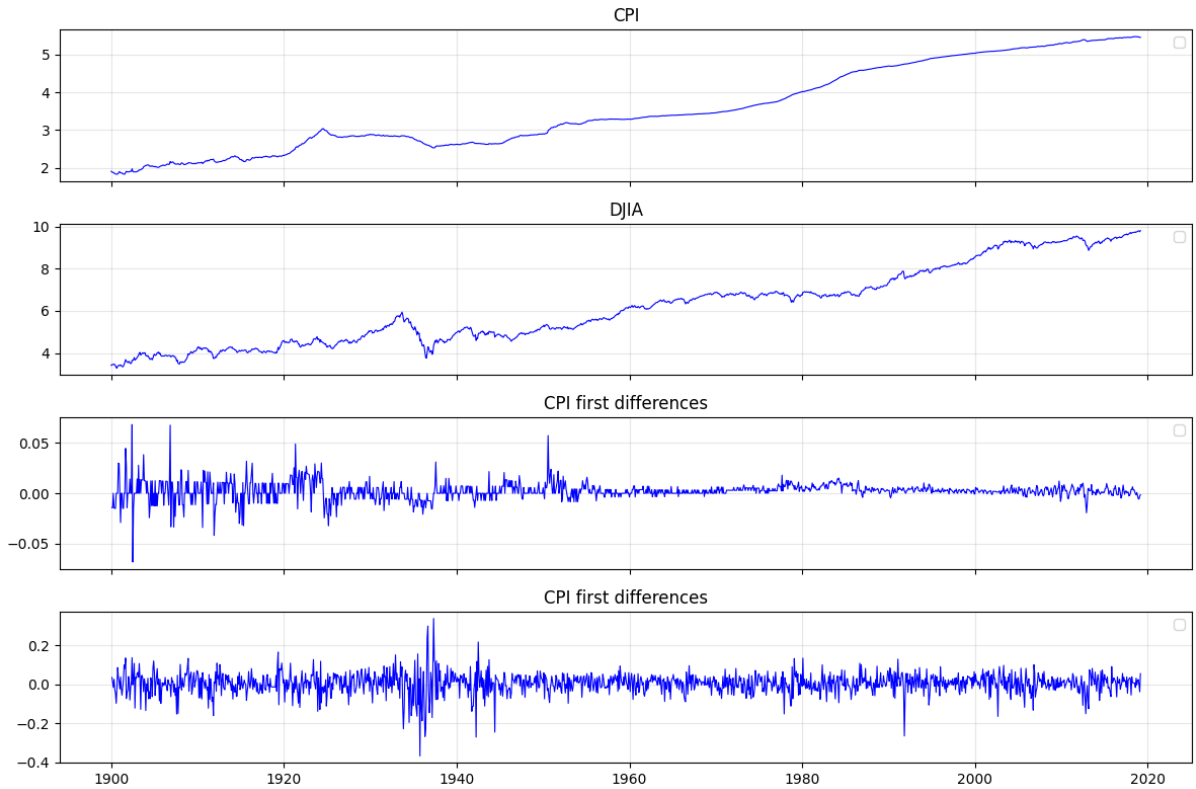

Figure 1: Data plots.

Table 1: Stationarity Tests (Task 1)

Series	ADF		KPSS	
	Statistic	p-value	Statistic	p-value
CPI (log level)	0.0038	0.9588	5.3745	0.010
DJIA (log level)	0.2806	0.9764	5.3225	0.010
π_t (d_CPI)	-5.8981	0.0000	0.2252	0.100
g_t (d_DJIA)	-8.8769	0.0000	0.0766	0.100

1.4 ARDL Model

The ARDL(p, q) model for inflation π_t with exogenous variable g_t is specified as:

$$\pi_t = \alpha + \sum_{i=1}^{p} \phi_i \pi_{t-i} + \sum_{j=0}^{q} \beta_j g_{t-j} + \varepsilon_t,$$

where ε_t is a white noise error term. The lag orders p and q are selected by minimising the Akaike Information Criterion (AIC) over a grid $p = 1, \ldots, 12$ and $q = 0, \ldots, 12$. Estimation is performed by ordinary least squares (OLS) after aligning the observations to avoid missing lags.

1.5 ARMAX Model

The ARMAX model combines an ARMA structure for inflation with the contemporaneous value of g_t as an exogenous regressor:

$$\pi_t = \alpha + \gamma g_t + \sum_{i=1}^{p} \varphi_i \pi_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t,$$

where ε_t is an innovation term. Orders p and q (for the AR and MA parts) are selected by AIC over $p, q = 0, \ldots, 3$. The model is estimated by maximum likelihood using the SARIMAX implementation in Python.

1.6 Diagnostics

The adequacy of each model is assessed by inspecting the autocorrelation function (ACF) of residuals and applying the Ljung-Box test for serial correlation up to lag 18. Additionally, information criteria (AIC, BIC) and the root mean squared error (RMSE) are used for model comparison.

1.7 ARDL Model results

The AIC-optimal ARDL specification is ARDL$(4, 3)$, i.e. four lags of inflation and three lags of stock market growth (including the contemporaneous value). The estimation results are reported in Table 2. The fitted equation is:

$$\begin{aligned} \hat{\pi}_t =& 0.0012 + 0.1861\pi_{t-1} + 0.1148\pi_{t-2} + 0.0676\pi_{t-3} + 0.1204\pi_{t-4} \\ &+ 0.0035 g_t + 0.0124 g_{t-1} + 0.0079 g_{t-2} + 0.0076 g_{t-3}. \end{aligned} \tag{1}$$

The contemporaneous effect of stock market growth is positive but statistically insignificant (p=0.380). However, the first and second lags of g_t are significant at the 5% level, and the third lag is marginally significant (p=0.060). The Ljung-Box test for the residuals at lag 18 yields a p-value of 0.0303, suggesting some remaining autocorrelation (rejection at 5%). The ACF plot (Figure 2) also reveals a few significant spikes, pointing to possible model misspecification.

Table 2: ARDL(4,3) Estimates

Variable	Coefficient	Std. Error	p-value
Constant	0.0012	0.0002	0.000
π_{t-1}	0.1861	0.0263	0.000
π_{t-2}	0.1148	0.0267	0.000
π_{t-3}	0.0676	0.0266	0.011
π_{t-4}	0.1204	0.0262	0.000
g_t	0.0035	0.0040	0.380
g_{t-1}	0.0124	0.0040	0.002
g_{t-2}	0.0079	0.0040	0.048
g_{t-3}	0.0076	0.0040	0.060
R^2		0.119	
Adjusted R^2		0.114	
AIC		-9707.36	
BIC		-9660.00	

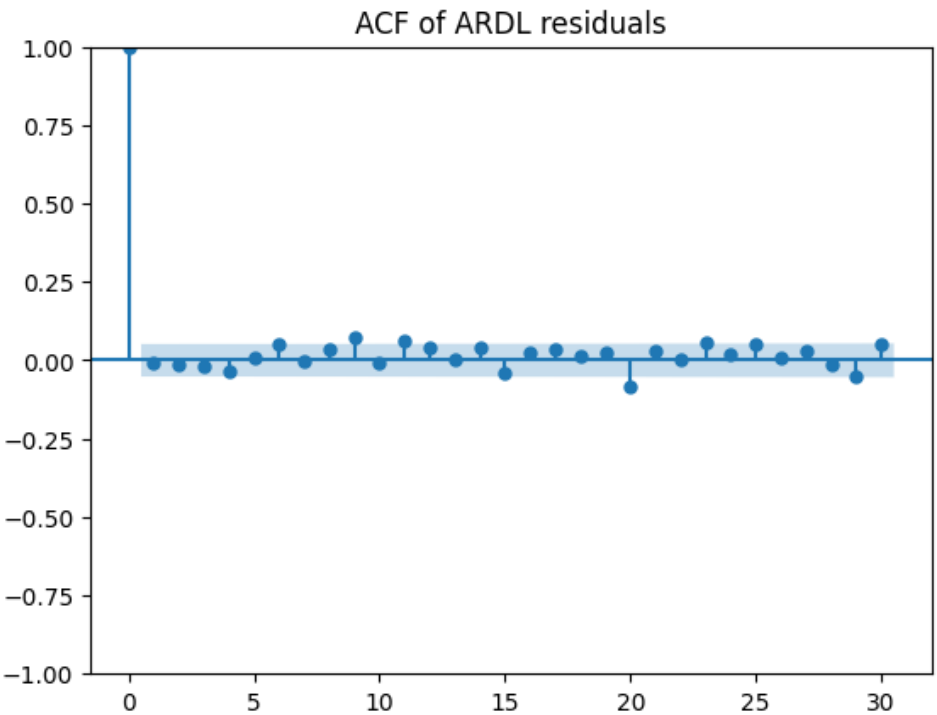

Figure 2: ACF of residuals from ARDL(4,3) model.

1.8 ARMAX Model results

The AIC-minimising ARMAX specification is ARMA$(1,1)$ with g_t as an exogenous variable. Estimation results are given in Table 3. The fitted equation is:

$$\hat{\pi}_t = 0.0003 - 0.0005 g_t + 0.8944 \pi_{t-1} - 0.7371 \varepsilon_{t-1} + \varepsilon_t. \tag{2}$$

Table 3: ARMAX(1,1) Estimates

Variable	Coefficient	Std. Error	p-value
Intercept	0.0003	7.23	0.000
g_t	-0.0005	0.0035	0.881
AR(1)	0.8944	0.0164	0.000
MA(1)	-0.7371	0.0205	0.000
σ^2	6.433	8.52	0.000
AIC		-9717.26	
BIC		-9690.94	

All coefficients except that of g_t are highly significant. The contemporaneous effect of g_t is negative but statistically insignificant (p=0.881). The Ljung-Box test at lag 18 gives a p-value of 0.4493, indicating no serial correlation in the residuals. The ACF plot (Figure 3) confirms that residuals resemble white noise.

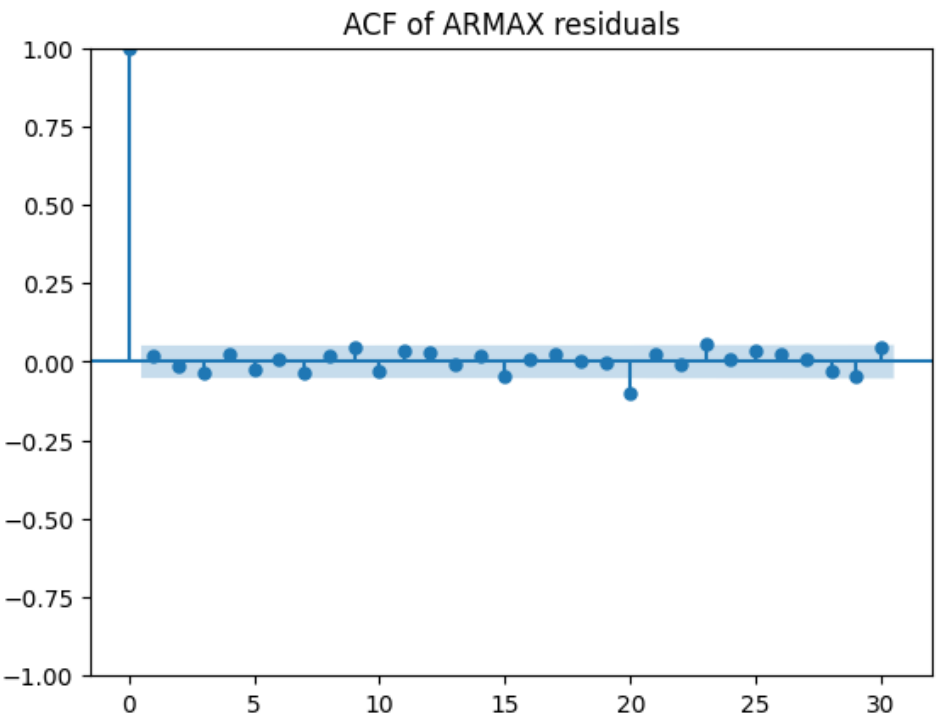

Figure 3: ACF of residuals from ARMAX(1,1) model.

1.9 Visual Comparison of Fitted Values

Figure 4 display the actual inflation series together with the fitted values from each model. Both models track the general movements of inflation reasonably well.

Figure 5 plots the residuals over time for both models. No obvious patterns or volatility clusters are visible, supporting the adequacy of the specifications.

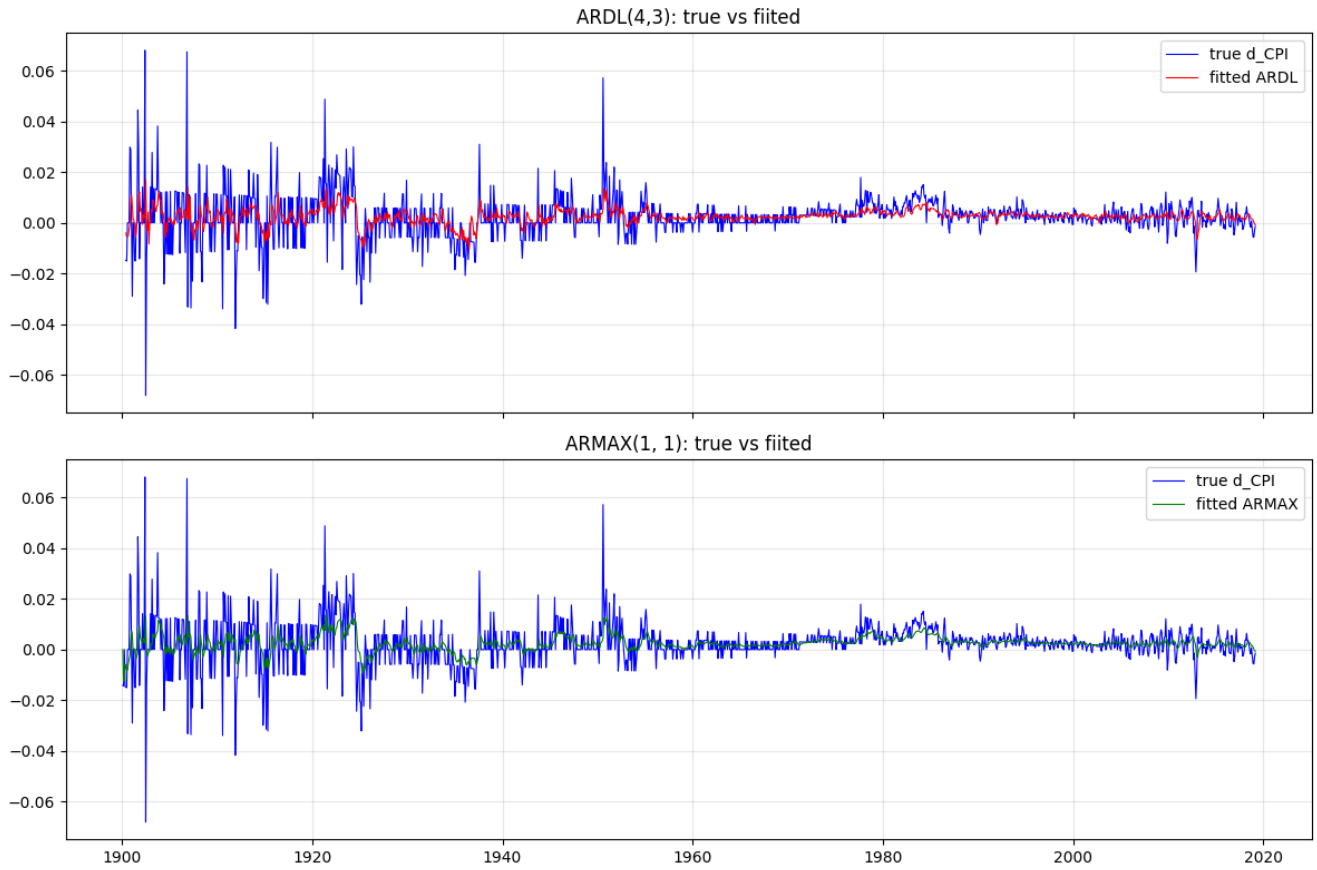

Figure 4: Actual vs fitted values – ARDL(4,3) and ARMAX(1,1).

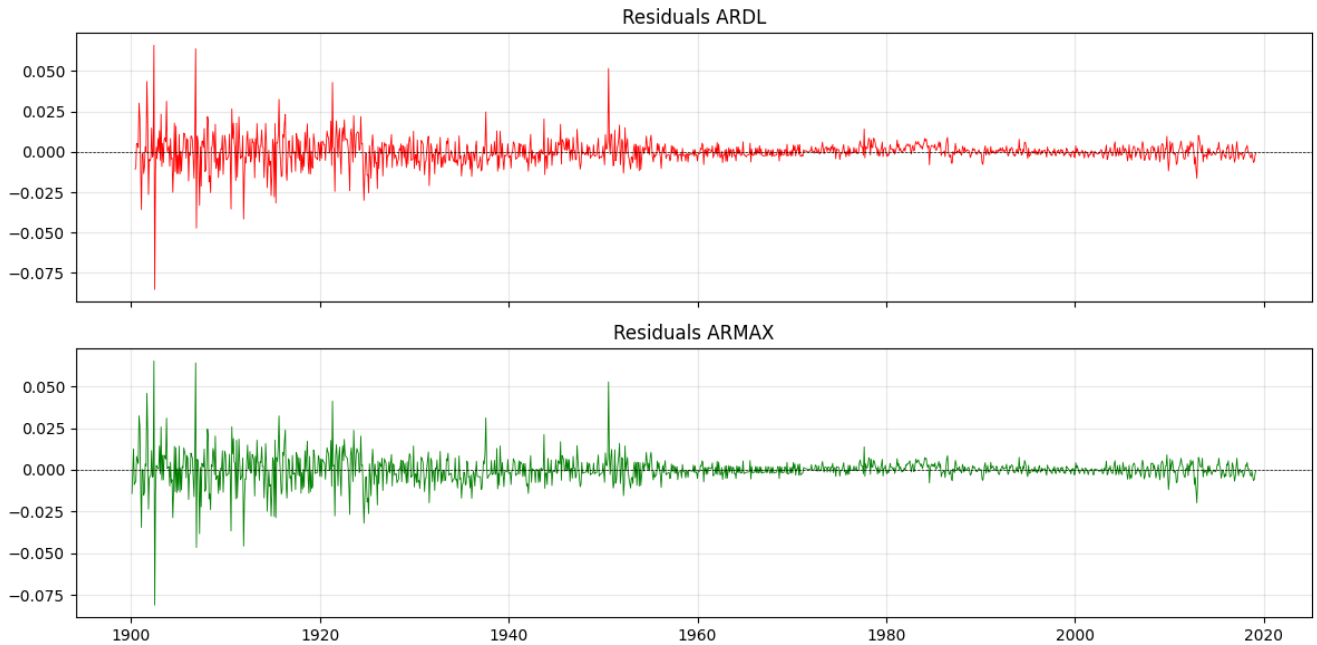

Figure 5: Residuals of ARDL (top) and ARMAX (bottom) models.

1.10 Model Comparison and Conclusion

Table 4 compares the two models in terms of information criteria and RMSE (computed on the estimation sample, with appropriate alignment for lags).

Table 4: Model Comparison

Model	AIC	BIC	RMSE
ARDL(4,3)	-9707.36	-9660.00	0.007995
ARMAX(1,1)	-9717.26	-9690.94	0.008027

The ARMAX model has slightly lower AIC and BIC, indicating a better balance of fit and parsimony. The RMSE values are very close, with ARDL marginally better. However, the ARMAX residuals satisfy the white noise assumption, whereas the ARDL residuals exhibit mild autocorrelation, favouring the ARMAX specification in terms of diagnostic checks.

We conclude that US inflation depends on its own past values but DJIA is not a significant predictor

2 Task 5

2.1 Data

The dataset contains monthly observations of the logarithm of the real Dow Jones Industrial Average (REALDJIA) from January 1900 to March 2019, giving 1431 observations. Let y_t denote the log index at time t. The growth rate is defined as $\Delta y_t = y_t - y_{t-1}$. After differencing, 1430 observations are available for modelling.

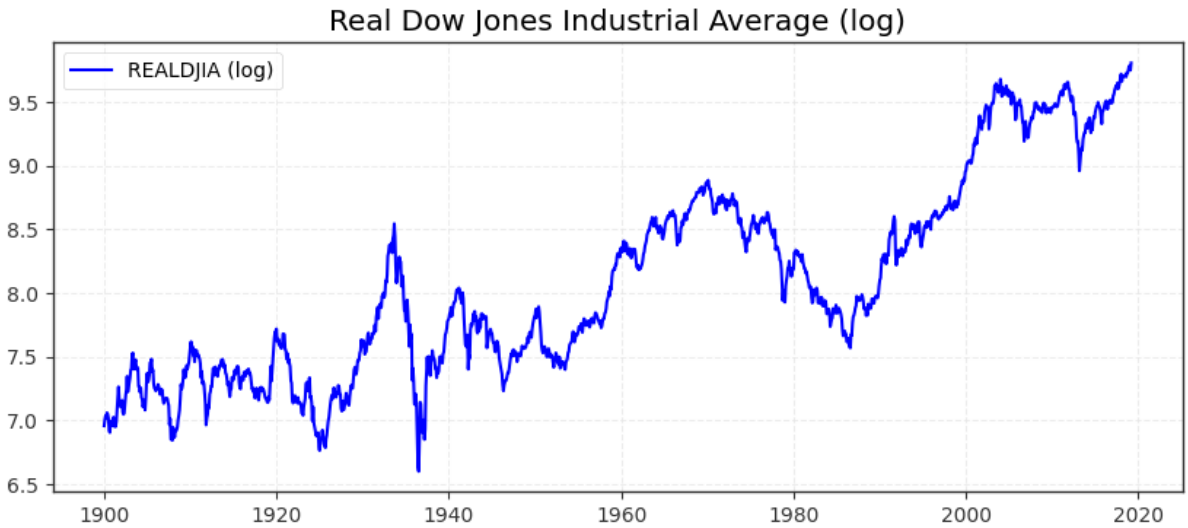

Figure 6: REALDJIA

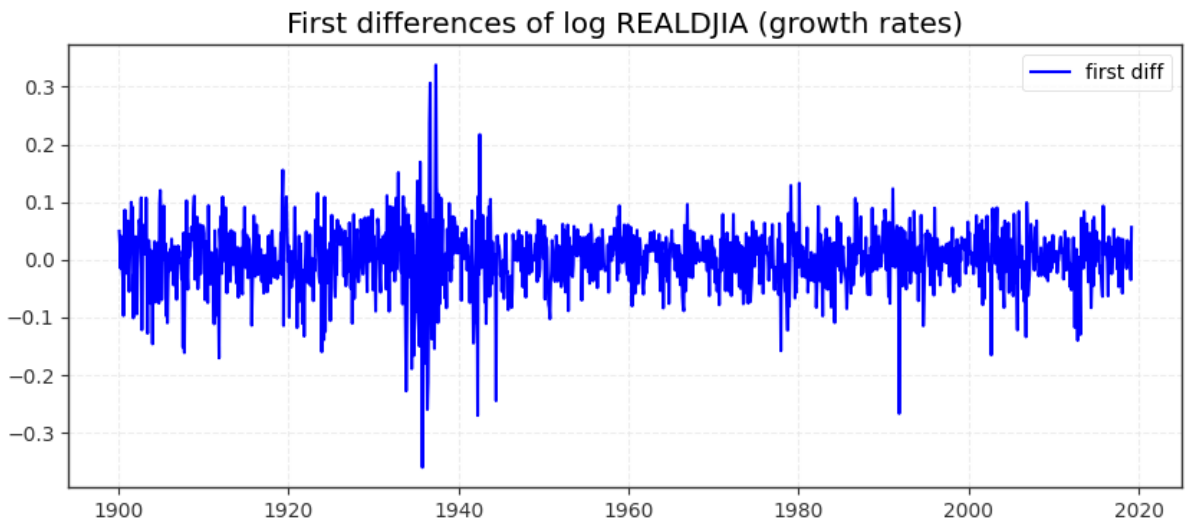

Figure 7: First Differences

2.2 Stationarity

Table 5 reports the test statistics. The log level is clearly non-stationary (ADF p-value > 0.05, KPSS rejects stationarity), while the first differences are stationary (ADF p-value < 0.01, KPSS does not reject). All subsequent modelling is therefore performed on Δy_t.

Table 5: Stationarity tests for REALDJIA

Series	ADF		KPSS	
	Statistic	p-value	Statistic	p-value
Log level	-0.7057	0.8453	4.4341	0.010
First difference	-8.9750	0.0000	0.0463	0.100

2.3 Model Estimates

The estimation results for the three models are summarised in Tables 6, 7 and 8. All coefficients are reported with standard errors and p-values.

2.4 Diagnostic Checks

The Ljung–Box test at lag 18 (Table 9) indicates that the residuals of the MA(14) model are white noise (p-value = 0.9999). The AR(5) model shows mild evidence of remaining autocorrelation (p-value = 0.1050, still above 0.05), while the ARMA(1,1) model is clearly inadequate (p-value = 0.0022). These findings are supported by the ACF plots of the residuals (not shown).

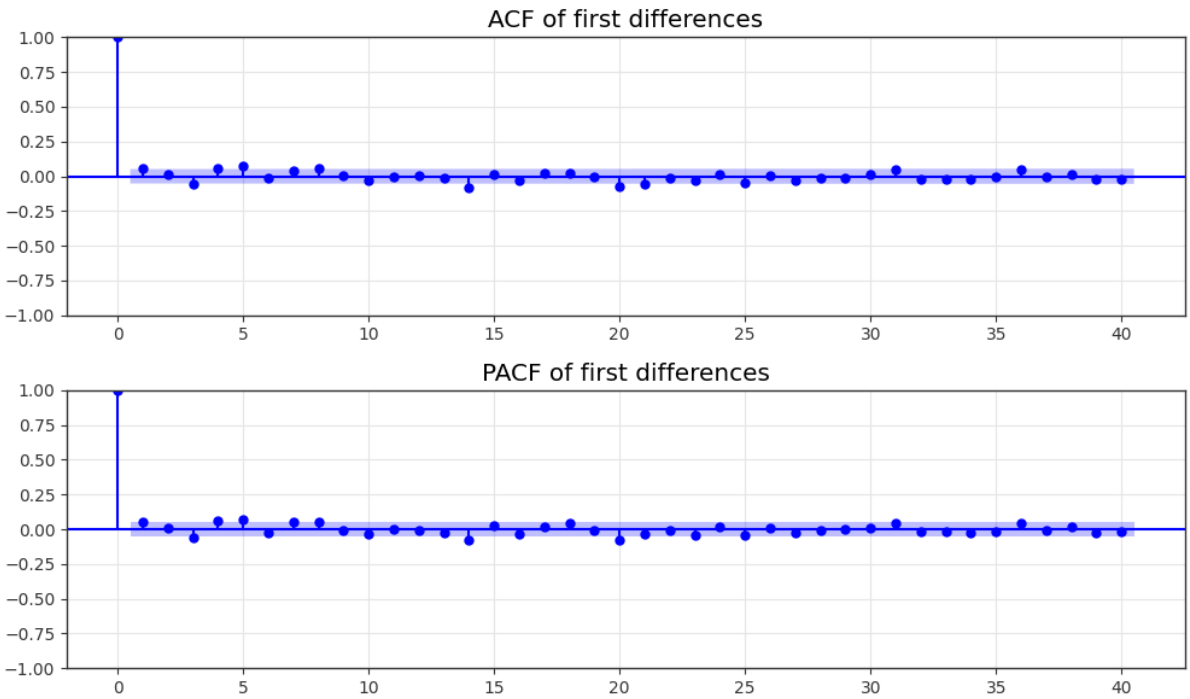

Figure 8: ACF and PACF of first differences

Table 6: AR(5) model for Δy_t

Variable	Coefficient	Std. Error	p-value
const	0.0020	0.0018	0.256
ar.L1	0.0561	0.0189	0.003
ar.L2	0.0156	0.0190	0.411
ar.L3	−0.0638	0.0163	0.000
ar.L4	0.0610	0.0192	0.002
ar.L5	0.0690	0.0202	0.001
σ^2	0.0029	6.39	0.000
AIC		-4302.42	
BIC		-4265.56	

2.5 Manual Calculation of MA(14) Forecasts for Periods 1 and 2

Model Specification

The estimated MA(14) model for the first differences y_t (growth rates) is

$$y_t = \mu + \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_{14}\varepsilon_{t-14},$$

where $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$. From the `statsmodels` output:

- Constant: $\mu = 0.0020$

Table 7: MA(14) model for Δy_t

Variable	Coefficient	Std. Error	p-value
const	0.0020	0.0017	0.234
ma.L1	0.0604	0.0210	0.004
ma.L2	0.0202	0.0194	0.299
ma.L3	−0.0685	0.0182	0.000
ma.L4	0.0574	0.0217	0.008
ma.L5	0.0836	0.0217	0.000
ma.L6	−0.0210	0.0199	0.292
ma.L7	0.0334	0.0197	0.091
ma.L8	0.0557	0.0174	0.001
ma.L9	0.0110	0.0190	0.561
ma.L10	−0.0446	0.0195	0.022
ma.L11	−0.0229	0.0195	0.239
ma.L12	0.0187	0.0210	0.372
ma.L13	−0.0149	0.0215	0.487
ma.L14	−0.0862	0.0222	0.000
σ^2	0.0028	6.62	0.000
AIC		-4307.55	
BIC		-4223.30	

Table 8: ARMA(1,1) model for Δy_t

Variable	Coefficient	Std. Error	p-value
const	0.0020	0.0015	0.201
ar.L1	0.0526	0.3285	0.873
ma.L1	0.0037	0.3274	0.991
σ^2	0.0029	6.06	0.000
AIC		-4290.50	
BIC		-4269.44	

Table 9: Ljung–Box test (lag 18) for model residuals

Model	p-value
AR(5)	0.1050
MA(14)	0.9999
ARMA(1,1)	0.0022

- MA coefficients:

$$\theta_1 = 0.0604, \quad \theta_2 = 0.0202, \quad \theta_3 = -0.0685,$$
$$\theta_4 = 0.0574, \quad \theta_5 = 0.0836, \quad \theta_6 = -0.0210,$$
$$\theta_7 = 0.0334, \quad \theta_8 = 0.0557, \quad \theta_9 = 0.0110,$$
$$\theta_{10} = -0.0446, \quad \theta_{11} = -0.0229, \quad \theta_{12} = 0.0187,$$
$$\theta_{13} = -0.0149, \quad \theta_{14} = -0.0862.$$

- Error variance: $\sigma^2 = 0.0028$.

Last 14 Residuals (chronological order, oldest to newest)

The residuals $\varepsilon_t, \varepsilon_{t-1}, \ldots, \varepsilon_{t-13}$ are:

Residual	Value
ε_{t-13}	0.05955442
ε_{t-12}	−0.03597968
ε_{t-11}	0.00425953
ε_{t-10}	0.03030392
ε_{t-9}	0.01226334
ε_{t-8}	−0.00233579
ε_{t-7}	0.03636211
ε_{t-6}	−0.02596343
ε_{t-5}	0.00265467
ε_{t-4}	0.01064203
ε_{t-3}	−0.00389009
ε_{t-2}	0.00093444
ε_{t-1}	0.03851327
ε_t	−0.05531734

(The list ends with −0.05531734, so we take that as ε_t, the most recent residual.)

1. One-Step Ahead Forecast ($h = 1$)

The optimal forecast for y_{t+1} is

$$\hat{y}_{t+1|t} = \mu + \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1} + \cdots + \theta_{14} \varepsilon_{t-13}.$$

Plugging in the numbers:

$$
\begin{aligned}
\hat{y}_{t+1|t} = 0.0020 & \\
& + 0.0604 \times (-0.05531734) \\
& + 0.0202 \times 0.03851327 \\
& - 0.0685 \times 0.00093444 \\
& + 0.0574 \times (-0.00389009) \\
& + 0.0836 \times 0.01064203 \\
& - 0.0210 \times 0.00265467 \\
& + 0.0334 \times (-0.02596343) \\
& + 0.0557 \times 0.03636211 \\
& + 0.0110 \times (-0.00233579) \\
& - 0.0446 \times 0.01226334 \\
& - 0.0229 \times 0.03030392 \\
& + 0.0187 \times 0.00425953 \\
& - 0.0149 \times (-0.03597968) \\
& - 0.0862 \times 0.05955442.
\end{aligned}
$$

Calculate each term (keeping sufficient precision):

Term	Product
$0.0604 \times (-0.05531734)$	-0.003340
0.0202×0.03851327	0.000778
$-0.0685 \times 0.00093444$	-0.000064
$0.0574 \times (-0.00389009)$	-0.000223
0.0836×0.01064203	0.000890
$-0.0210 \times 0.00265467$	-0.000056
$0.0334 \times (-0.02596343)$	-0.000867
0.0557×0.03636211	0.002025
$0.0110 \times (-0.00233579)$	-0.000026
$-0.0446 \times 0.01226334$	-0.000547
$-0.0229 \times 0.03030392$	-0.000694
0.0187×0.00425953	0.000080
$-0.0149 \times (-0.03597968)$	0.000536
$-0.0862 \times 0.05955442$	-0.005134

Sum of all products:

$$\begin{aligned}\sum \theta_i \varepsilon_{\dots} &\approx (-0.003340) + 0.000778 - 0.000064 - 0.000223 + 0.000890 \\ &\quad - 0.000056 - 0.000867 + 0.002025 - 0.000026 - 0.000547 \\ &\quad - 0.000694 + 0.000080 + 0.000536 - 0.005134 \\ &= -0.007642.\end{aligned}$$

Finally add the constant $\mu = 0.0020$:

$$\hat{y}_{t+1|t} = 0.0020 - 0.007642 = \boxed{-0.005642}.$$

Forecast Error Variance and Confidence Intervals For $h = 1$, the forecast error variance is simply $\sigma^2 = 0.0028$. Standard error: $\sqrt{0.0028} = 0.052915$.

95% confidence interval: $\hat{y}_{t+1|t} \pm 1.96 \times 0.052915$ $-0.005642 \pm 0.103715 \Rightarrow (-0.109357,\ 0.098073)$

2. Two-Step Ahead Forecast ($h = 2$)

For $h = 2$, the optimal forecast conditions on information up to time t:

$$\hat{y}_{t+2|t} = \mu + \theta_2 \varepsilon_t + \theta_3 \varepsilon_{t-1} + \cdots + \theta_{14} \varepsilon_{t-12}.$$

(Future errors ε_{t+1} are unknown and replaced by zero.)

Plug in the numbers – note that the terms now involve ε_t down to ε_{t-12}:

$$\begin{aligned}\hat{y}_{t+2|t} &= 0.0020 \\ &\quad + 0.0202 \times (-0.05531734) \\ &\quad - 0.0685 \times 0.03851327 \\ &\quad + 0.0574 \times 0.00093444 \\ &\quad + 0.0836 \times (-0.00389009) \\ &\quad - 0.0210 \times 0.01064203 \\ &\quad + 0.0334 \times 0.00265467 \\ &\quad + 0.0557 \times (-0.02596343) \\ &\quad + 0.0110 \times 0.03636211 \\ &\quad - 0.0446 \times (-0.00233579) \\ &\quad - 0.0229 \times 0.01226334 \\ &\quad + 0.0187 \times 0.03030392 \\ &\quad - 0.0149 \times 0.00425953 \\ &\quad - 0.0862 \times (-0.03597968).\end{aligned}$$

Compute each product:

Term	Product
$0.0202 \times (-0.05531734)$	-0.001117
$-0.0685 \times 0.03851327$	-0.002638
0.0574×0.00093444	0.000054
$0.0836 \times (-0.00389009)$	-0.000325
$-0.0210 \times 0.01064203$	-0.000223
0.0334×0.00265467	0.000089
$0.0557 \times (-0.02596343)$	-0.001446
0.0110×0.03636211	0.000400
$-0.0446 \times (-0.00233579)$	0.000104
$-0.0229 \times 0.01226334$	-0.000281
0.0187×0.03030392	0.000567
$-0.0149 \times 0.00425953$	-0.000063
$-0.0862 \times (-0.03597968)$	0.003102

Sum the products:

$$\begin{aligned}\sum &\approx (-0.001117) - 0.002638 + 0.000054 - 0.000325 - 0.000223 \\ &\quad + 0.000089 - 0.001446 + 0.000400 + 0.000104 - 0.000281 \\ &\quad + 0.000567 - 0.000063 + 0.003102 \\ &= -0.001777.\end{aligned}$$

Add constant $\mu = 0.0020$:

$$\hat{y}_{t+2|t} = 0.0020 - 0.001777 = \boxed{0.000223}.$$

Forecast Error Variance and Confidence Intervals For an MA(q) process, the h-step forecast error variance is

$$\mathrm{Var}(e_{t+h|t}) = \sigma^2 \left(1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_{h-1}^2\right), \quad h \leq q.$$

Here $q = 14$, $h = 2$:

$$\mathrm{Var}(e_{t+2|t}) = \sigma^2 \left(1 + \theta_1^2\right).$$

- $\theta_1^2 = (0.0604)^2 = 0.003648$
- $1 + \theta_1^2 = 1.003648$
- Variance $= 0.0028 \times 1.003648 = 0.002810$

- Standard error $= \sqrt{0.002810} = 0.053014$

Now compute intervals:

- **95% CI:** $0.000223 \pm 1.96 \times 0.053014$ $0.000223 \pm 0.103908 \Rightarrow (-0.103685,\ 0.104131)$

Summary Table

Horizon	Point Forecast	95% CI
1	-0.005642	$(-0.109357,\ 0.098073)$
2	0.000223	$(-0.103685,\ 0.104131)$

2.6 Forecast

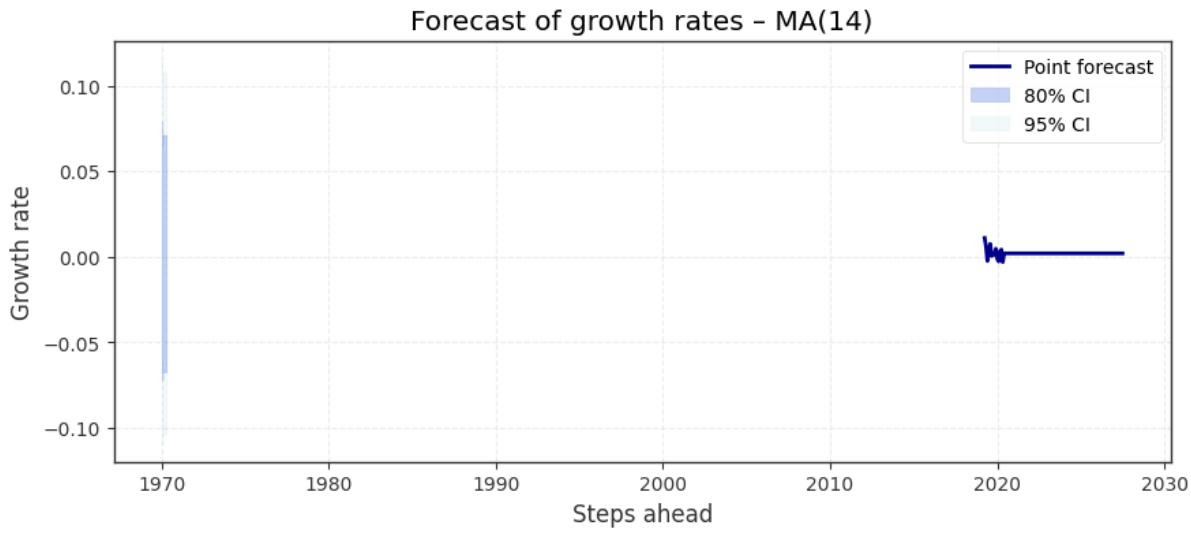

Figure 9: MA(14) forecast

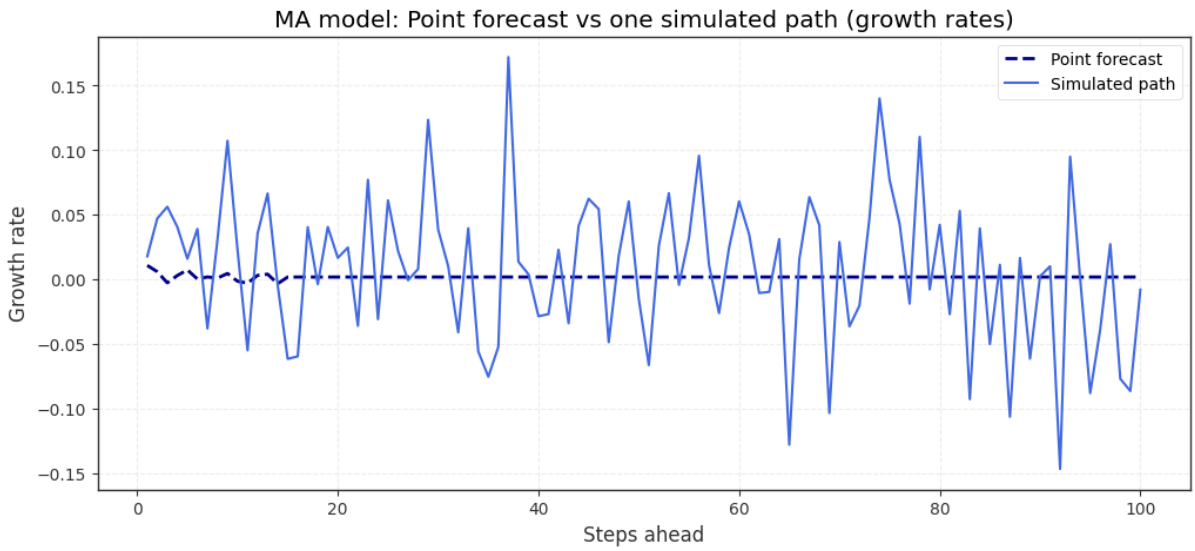

Figure 10: Simulation